

JAY RAO

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EDUCATION

Arizona State University

Aug 2025 - May 2029

B.S.E., Electrical Engineering

- GPA: 3.51

ACADEMIC PROJECTS

Chief Technical Officer

May 2026 - Present

Tack Braille LLC | ASU, Tempe

- Co-founded and lead full technical direction for Tack, a screen-reader-first AI web assistant for blind and visually impaired users; originated as an ASU EPICS braille-display hardware prototype (Jan-May 2026) before incorporating as Tack Braille LLC and pivoting the product to software, targeting public launch by end of 2026.
- Own architecture and delivery for a Next.js/TypeScript webapp with a security- and accessibility-hardened backend (auth, rate limiting, SSRF/IDOR guards, WCAG AA compliance); direct engineering workflow and manage cross-timezone partner and vendor relationships

RC Car Autonomous Robot

Jun 2026 - Present

Personal Build | Tempe, AZ

- Building a two-module hardware setup: Raspberry Pi 4 for computer vision and neural network inference, RP2040 for real-time motor control via L298N H-bridge; wired the BLE sensor pipeline and verified drive and steering motor control (Phases 1-2 of 8)
- Building USB serial command protocol between Pi 4 and RP2040 (Phase 3); roadmap includes OpenCV obstacle detection, biomimicry-based neural network training, and digital twin simulation for autonomous navigation

Self-Balancing PID Robot

Jun 2026 - Present

Personal Build | Tempe, AZ

- Designing the hardware and control stack: MPU6050 IMU over I2C on RP2040, complementary filter for gyro and accelerometer pitch fusion, PID control loop targeting disturbance recovery within 2 seconds; shares L298N motor driver and RP2040 with RC Car build
- System designed and ready to assemble; awaiting MPU6050 sensor; next steps include I2C verification, calibration sketch, filter implementation, and iterative PID tuning starting with proportional gain only

SpaceHACK for Sustainability

Mar 2026

W.P. Carey School of Business | Tempe, AZ

- Placed 1st at ASU and 2nd internationally at SpaceHACK for Sustainability 2026, competing against over 150 teams from universities across the globe
- Leveraged Google Earth Engine, Sentinel-5P CO datasets, SRTM elevation data, and live vessel tracking to identify the most efficient overland corridor

TECHNICAL SKILLS

Hardware & Embedded (learning): RP2040, Raspberry Pi 4, L298N H-bridge, I2C, PWM, Motor Control, MATLAB

Programming: C++, Python, MicroPython, Arduino IDE, OpenCV (learning)

Software and Tools: SolidWorks, MATLAB, AutoCAD, Git, Microsoft Office, Excel

VOLUNTEERING EXPERIENCE

Lead Volunteer

Sep 2023 - Jun 2025

Shiva Vishnu Temple, Livermore, CA

- Designed and built a new cultural education room serving 120+ students and families; managed layout, material sourcing, and a small volunteer team through completion
- Created digital learning modules (Python/Arduino basics) to support youth programming

Mentor

Apr 2024 - Jun 2025

Arduino Club

- Taught foundational Arduino concepts (loops, input/output, sensor reading)
- Mentored 15 freshmen, guiding them through building simple circuits and microcontroller projects

JAY RAO - MASTER COURSE CATALOG

Reference source for tailoring - not a direct application document.

FALL 2026 - IN PROGRESS

EEE 202 - Circuits I

Status: In Progress

- Lecture/laboratory coursework covers circuit elements, Ohm's and Kirchhoff's laws, nodal and mesh analysis, network theorems, operational amplifiers, DC/AC steady-state behavior, first- and second-order transients, phasors, impedance, transfer functions, filters, and Bode plots. Labs combine SPICE/LTspice and MATLAB simulation with circuit construction and measurement using an oscilloscope/waveform generator, digital multimeter, and discrete components.

MAT 267 - Calculus for Engineers III

Status: In Progress

- Multivariable and vector calculus for engineering applications, including vectors and space curves, partial and directional derivatives, gradients and optimization, double and triple integrals, vector fields, line and surface integrals, Green's theorem, curl, and divergence.

MAT 275 - Modern Differential Equations

Status: In Progress

- Applied ordinary differential equations, initial-value problems, direction fields, stability, numerical methods, characteristic equations, damped and forced vibrations, Laplace transforms, piecewise and impulse forcing, and eigenvalue-based linear systems, with MATLAB used for modeling and numerical solutions.

PHY 131 - University Physics II: Electricity and Magnetism

Status: In Progress

- Calculus-based lecture and recitation covering electric charge and current, electric and magnetic fields in vacuum and matter, induction, DC and AC circuits, displacement current, and electromagnetic waves, with recitation reinforcing quantitative multi-step problem solving.

PHY 132 - University Physics Laboratory II

Status: In Progress

- Calculus-based electricity and magnetism laboratory involving circuit construction, RC/RLC behavior, magnetic fields, and induction. Coursework develops precise measurement with voltmeters, ammeters, and oscilloscopes, experimental data analysis, mathematical modeling, and technical lab-report writing.

GPH 111 - Introduction to Physical Geography

Status: In Progress

- Examines spatial and functional relationships among climate, landforms, soils, water, and vegetation, emphasizing how interacting physical systems shape environments across regions.

SPRING 2026 - DOCUMENTED COURSEWORK

CSE 100 - Principles of Programming with C++

Status: Enrollment Documented

- Lecture/laboratory introduction to structured problem solving in C++. Coursework covers algorithm design, structured programming, fundamental algorithms and techniques, computer-systems concepts, and social and ethical responsibility; programming labs emphasize implementing, testing, and debugging C++ solutions.

EEE 120 - Digital Design Fundamentals

Status: Enrollment Documented

- Lecture/laboratory study of number systems, binary and complement arithmetic, Boolean algebra, logic minimization, ROMs and PLAs, flip-flops, and synchronous sequential circuits. Coursework includes designing, simulating, building, and debugging synchronous machines and an elementary microprocessor.

ENG 102 - First-Year Composition

Status: Enrollment Documented

- Research-focused academic writing and argumentation, including evidence discovery, claim support, competing perspectives, source evaluation and documentation, drafting, collaboration, revision, and reflection. ASU's 15-week course includes three formal writing projects totaling approximately 5,000 words.

FSE 104 - EPICS Gold I

Status: Enrollment Documented

- Human-centered engineering design and entrepreneurial thinking applied to a team project for a community partner. Coursework develops needs discovery, iterative prototyping, teamwork, communication, and design/build/implementation deliverables with real-world community impact.

MAT 266 - Calculus for Engineers II

Status: Enrollment Documented

- Engineering calculus covering integration methods and applications, numerical and improper integration, areas, volumes, arc length, and work; sequences and convergence; power, Taylor, and Maclaurin series; and parametric and polar calculus.

PHY 121 - University Physics I: Mechanics

Status: Enrollment Documented

- Calculus-based Newtonian mechanics covering vectors, linear and rotational kinematics and dynamics, Newton's laws, forces, work, energy, momentum, collisions, conservation laws, gravitation, and oscillatory motion, with emphasis on quantitative modeling and multi-step problem solving.

PHY 122 - University Physics Laboratory I

Status: Enrollment Documented

- Calculus-based mechanics laboratory with experiments involving motion, vectors and statics, Newton's laws, friction, energy and momentum, collisions, rotational motion, Hooke's law, waves, and sound. Coursework develops experimental measurement, data analysis, mathematical modeling, and formal lab-report writing.

EVIDENCE STATUS POLICY

- Calendar-confirmed enrollment is transferable context, not proof of proficiency.
- User-provided historical course lists remain qualified as documented coursework.
- A self-reported source-resume skill may remain in the skills section, but it cannot support an experience bullet without project or work evidence.
- ACE units without cold evidence are unsupported.
- Roadmap, planned, and target outcomes may remain visibly qualified in source material but cannot support a job-description claim.